

Spring 2025

GRADE 4
MATHEMATICS

M18_E665_2

1. Oscar started with 14 meters of ribbon. He cut off and used a piece of ribbon that was $9\frac{1}{4}$ meters long. What was the length of the ribbon he had left?
- A. $4\frac{1}{4}$ meters
- B. $4\frac{3}{4}$ meters
- C. $5\frac{1}{4}$ meters
- D. $5\frac{3}{4}$ meters

PTM0408_200

2. To find the product of 75 and 43, find the sum of the partial products $15 + 210 + \underline{\hspace{1cm}}? + 2,800$ to get 3,225.

What is the value of the missing partial product?

Enter your answer in the space.

PTM0427

- 3.** Greendale Elementary School students are taking a field trip. There are 6 classes with 25 students each and 1 class with 22 students. Each bus can hold 45 students.
- Write an equation to represent the number of buses needed for the field trip, x .
 - How many buses will the school need for the field trip?
 - How did you interpret the remainder to find the number of buses the school needs?

Enter your answer in the space provided. Show all your work to support your answer.

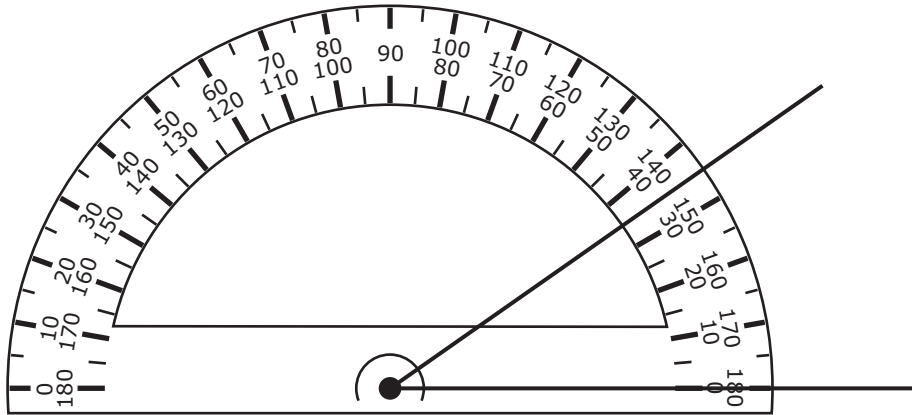
PTM0407_5523

- 4.** What is the sum of 3,054 and 2,469?

Enter your answer in the box.

PTM0404_1

5. An angle is shown with a protractor.



What is the measure of the angle, in degrees?

- A. 35°
- B. 45°
- C. 145°
- D. 155°

PTM0414

6. Hector wants to find the sum of $\frac{4}{10}$ and $\frac{5}{100}$.

Hector adds 4 and 5 and says the answer is $\frac{9}{100}$.

There is an error in Hector's work.

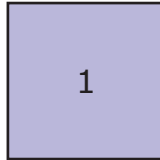
- Explain the error Hector made.
- Find the correct answer.
- Show work or explain how to determine the answer.

Enter your answer and your explanation in the space.

M18_B031_2

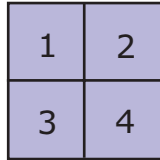
7. Evelyn made a pattern using squares. She started with 1 square. Each figure that follows has one more row of squares and one more column of squares than the previous figure.

Figure 1



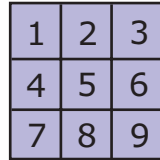
1 square

Figure 2



4 squares

Figure 3



9 squares

How many squares will Evelyn need to form Figure 6 in the pattern?

- A. 49
- B. 36
- C. 25
- D. 16

PTM0417_8:3

8. This question has two parts.

A camp counselor at a multicultural summer camp has 75 granola bars that she wants to share evenly among the 9 children in her group. She plans to keep any that are left over for herself.

Part A

How many granola bars will each child receive?

Enter your answer in the space.

Part B

How many granola bars will the camp counselor keep?

Enter your answer in the space.

PTM0416_3

9. A chef served 359 portions of grilled salmon at the Native American Food Festival. Each portion of salmon was 4 ounces.

How many ounces of salmon did the chef serve?

- A. 1,077 ounces
- B. 1,406 ounces
- C. 1,436 ounces
- D. 1,795 ounces

PTM0402_72

- 10.** What is the number of inches in 6 feet?

Enter your answer in the space.

PTM0411_P_5,1

- 11.** Compare the fractions.

Select **two** correct answers.

A. $\frac{3}{5} < \frac{5}{6}$

B. $\frac{3}{5} > \frac{5}{6}$

C. $\frac{3}{5} = \frac{5}{6}$

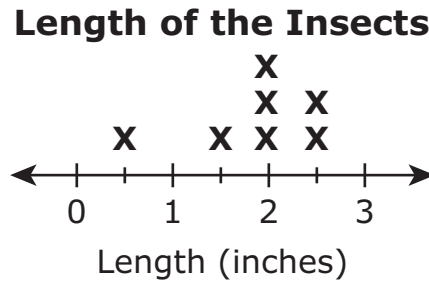
D. $\frac{3}{8} < \frac{1}{4}$

E. $\frac{3}{8} > \frac{1}{4}$

F. $\frac{3}{8} = \frac{1}{4}$

PTM0423_2

- 12.** Sam starts an insect collection. The length of each insect is shown in the line plot in inches.



What is the difference in length in inches between the longest and shortest insects in Sam's collection?

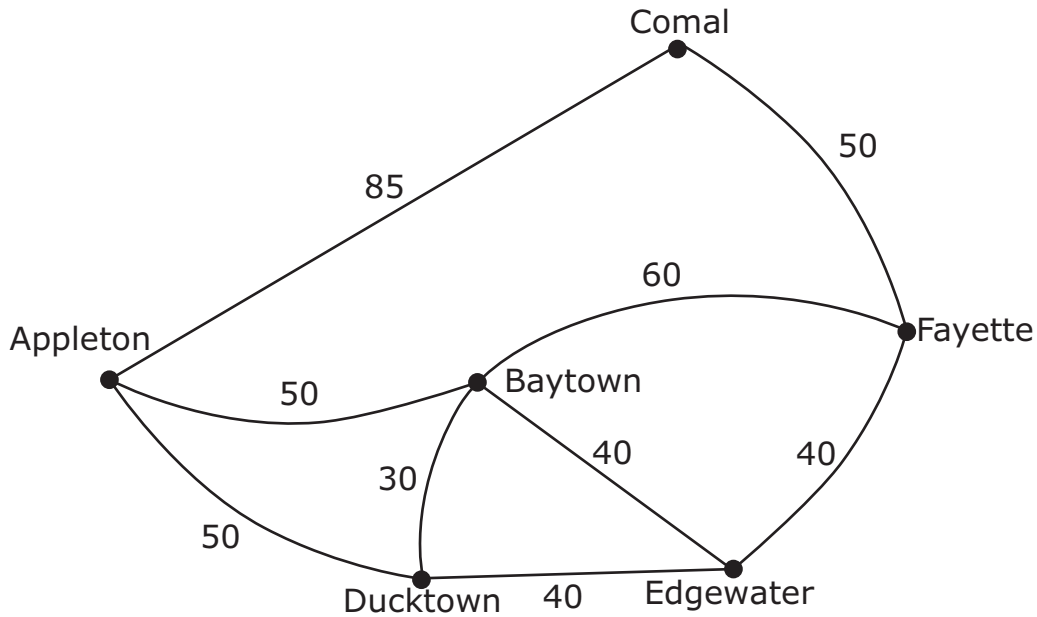
Enter your answer in the space.

PTM0405_2

- 13.** How does the digit 6 in the number 56,819 relate to the digit 6 in the number 467?
- A.** It is 10 times as great.
 - B.** It is 100 times as great.
 - C.** It is 10 times less.
 - D.** It is 100 times less.

M18_B184_4

- 14.** The figure shows the locations of the towns of Appleton, Baytown, Comal, Ducktown, Edgewater, and Fayette and the roads between them. The distances shown are in miles.



Which of these is the shortest route from Comal to Ducktown?

- A.** Comal to Appleton to Ducktown
- B.** Comal to Fayette to Baytown to Ducktown
- C.** Comal to Appleton to Baytown to Ducktown
- D.** Comal to Fayette to Edgewater to Ducktown

PTM0420

- 15.** There are 12 girls and boys at a quinceañera, where $\frac{3}{4}$ of them are girls and $\frac{1}{4}$ are boys. Each girl ate $\frac{1}{4}$ of a pizza, and each boy ate $\frac{1}{2}$ of a pizza.

- Show how to find the number of girls and boys at the quinceañera.
- Show how to find the amount of pizza the girls ate and the amount of pizza the boys ate.
- Did the girls or boys eat more pizza? How do you know?

Enter your answer in the space provided. Show all your work to support your answer.

PTM0428_:

- 16.** This question has two parts.

Part A

Some students were comparing the fractions $\frac{2}{4}$ and $\frac{1}{2}$.

Jace claimed that $\frac{2}{4}$ is greater than $\frac{1}{2}$ because 2 is greater than 1.

- Explain why Jace's reasoning is incorrect.
- What is the correct comparison of $\frac{2}{4}$ and $\frac{1}{2}$?

Enter your explanation and your answer in the space provided.

Part B

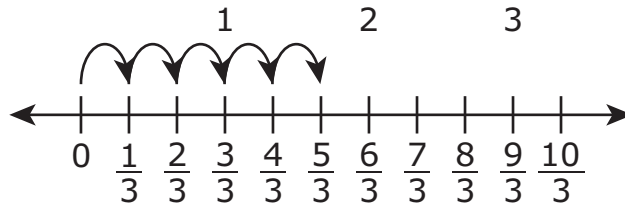
The students were also comparing the fractions $\frac{3}{5}$ and $\frac{3}{7}$. Catherine claimed that $\frac{3}{5}$ is less than $\frac{3}{7}$ because 5 is less than 7.

- Explain why Catherine's reasoning is incorrect.
- What is the correct comparison of $\frac{3}{5}$ and $\frac{3}{7}$?

Enter your answer and your explanation in the space provided.

PTM0413_1

17. What expression is represented by the number line?



- A. $\frac{1}{3} \times 5$
- B. $\frac{1}{3} \times 4$
- C. $5 \times \frac{1}{2}$
- D. $4 \times \frac{1}{2}$

PTM0406_3

18. The number of people in attendance at the German Heritage Festival was 18,769.

What is the number of attendees rounded to the nearest hundred?

- A. 20,000
- B. 19,000
- C. 18,800
- D. 18,770

PTM0421_:

19. Part A

Every morning, a restaurant worker stocks 8 shelves with beans. She puts 3 cans of pinto beans and 8 cans of black beans on each shelf.

- Write an equation that can be used to find the total number of cans of beans the restaurant worker stocks on shelves each morning.
- What is the total number of cans of beans?
- Show your work or explain how you determined your answer.

Enter your answer and your work in the space.

Part B

Every morning, the restaurant worker also stocks 6 shelves with rice. She puts 7 bags of white rice and 3 bags of brown rice on each shelf.

- Write an equation that can be used to find the total number of bags of rice the restaurant worker stocks on shelves each morning.
- What is the total number of bags of rice?
- Show your work or explain how you determined your answer.

Enter your answer and your work in the space.

PTM0412_2:3

20. This question has two parts.

Devon bought a replica model of the Golden Gate Bridge. He used three different colors of paint to paint the model. He used $\frac{3}{8}$ gallon of red paint, $\frac{7}{8}$ gallon of blue paint, and $\frac{2}{8}$ gallon of green paint.

Part A

How much red paint and green paint did Devon use?

- A. $\frac{1}{8}$ gallon
- B. $\frac{5}{8}$ gallon
- C. $\frac{9}{8}$ gallons
- D. $\frac{10}{8}$ gallons

Part B

How much more blue paint did Devon use than green paint?

- A. $\frac{10}{8}$ gallons
- B. $\frac{9}{8}$ gallons
- C. $\frac{5}{8}$ gallon
- D. $\frac{4}{8}$ gallon

PTM0418

21. Olivia has \$20. She earns \$12 each hour for doing chores. She does chores for 4 hours. Olivia wants to buy as many \$8 shirts as she can.

- Write an equation to find the number of \$8 shirts Olivia can buy, s .
- How many \$8 shirts can Olivia buy?
- How did you interpret the remainder to find the number of \$8 shirts Olivia can buy?

Enter your answer in the space provided. Show all your work to support your answer.

PTM0424_4

22. Which statement about angles is true?

- A.** Angles are always between 0 and 90 degrees.
- B.** Angles are formed whenever three rays share a common endpoint.
- C.** Angles that turn through $\frac{1}{2}$ of a circle are called “one-degree angles.”
- D.** Angles are formed whenever two rays share a common endpoint.

PTM0403_22

23. Kyle builds a rectangular vegetable planter that is 45 inches long and has a perimeter of 134 inches.

How many inches wide is Kyle’s vegetable planter?

Enter your answer in the space.

PTM0409_20

- 24.** To find the quotient of 984 divided by 8, find the sum of the partial quotients shown to get 123.

$$\begin{array}{r} 3 \\ \boxed{?} \\ 100 \\ 8 \overline{)984} \\ \underline{-800} \\ 184 \\ \underline{-160} \\ 24 \\ \underline{-24} \\ 0 \end{array}$$

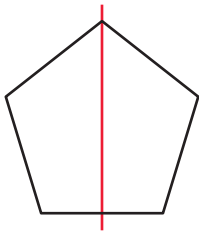
What is the value of the missing partial quotient?

Enter your answer in the space.

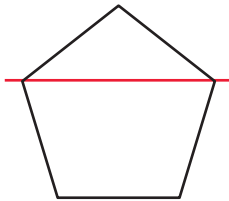
PTM0401_1

25. Which shape includes a line of symmetry?

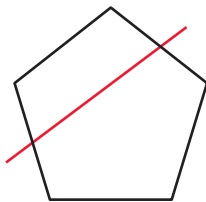
A.



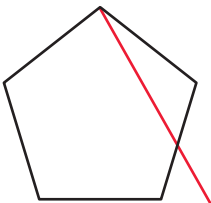
B.



C.



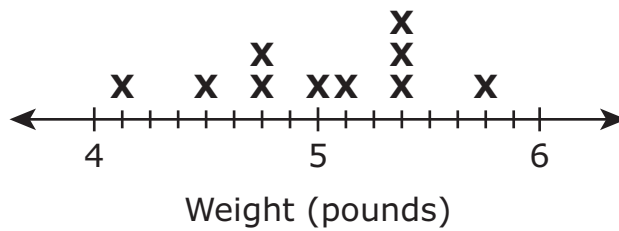
D.



4M23_090_2

- 26.** The line plot shows the weights, in pounds, of 10 baskets of berries for sale at a farm stand.

Basket Weights



KEY

Each **X** represents 1 basket.

Rohan bought the 6 heaviest baskets of berries. What is the total weight of all the baskets of berries Rohan bought?

- A.** $32\frac{3}{8}$ pounds
- B.** 32 pounds
- C.** $30\frac{4}{8}$ pounds
- D.** 27 pounds

PTM0415_3,1

27. Which comparisons are true?

Select **two** correct answers.

A. $1.2 > 0.18$

B. $0.9 < 0.09$

C. $0.04 < 0.15$

D. $0.37 > 0.4$

E. $5.86 < 5.09$

Illinois MATH Assessment

Practice Item Answer Key

Grade 4 - Paper, Screen Reader, and Non-Screen Reader

The following pages include the answer key for all machine-scored items, followed by a sample response for the hand-scored item.

- The rubrics show sample student responses. Student responses other than that shown in the rubric may earn full or partial credit.
- Which responses to hand-scored items receive full or partial credit will be confirmed during range-finding (reviewing sets of real student work)
- If students make a computation error, they can still earn points for reasoning or modeling.

Item Number	Answer Key
1.	B
2.	200
3.	See Rubric
4.	5523
5.	A
6.	See Rubric
7.	B
8.	See Rubric
9.	C
10.	72
11.	E, A
12.	2
13.	B
14.	D
15.	See Rubric
16.	See Rubric



17.	A
18.	C
19.	See Rubric
20.	Part A: B Part B: C
21.	See Rubric
22.	D
23.	22
24.	20
25.	A
26.	B
27.	C, A

3
Rubric

3 Point Constructed Response Rubric

Score	Description
3	Student response includes the following elements. • Reasoning/Modeling component = 1 point: Correct equation to find the number of buses needed for the field trip, x. • Reasoning/Modeling component = 1 point: Correct number of buses needed for the field trip. • Computation component = 1 point: Correct interpretation of the remainder. Sample Student Response: The equation used to find the number of school buses needed is: $(6 \times 25 + 22) \div 45 = x$ To find the number of school buses needed, solve the equation. To find the number of school buses needed, first you need to find the total number of students by multiplying 6 by 25, then adding 22. Then, divide by the number of students each school bus holds, 45. $(6 \times 25 + 22) \div 45 = x$ $(150 + 22) \div 45 = x$



	$172 \div 45 = x$ When dividing 172 by 45, 45 goes in to 172 3 times because 45 times 3 equals 135. Then, there are 37 students left over. 172 divided by 45 is 3 R 37. The school will need 4 buses, because when dividing, there are 37 students that will not fit on the third bus. They will need one more bus to hold the remaining students. Or other valid approaches are acceptable.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

# 6 Rubric	
3 Point Constructed Response Rubric	
Score	Description
3	Student response includes the following elements. • Reasoning/Modeling component = 1 point: Explains Hector's error • Computation component = 1 point: Correct sum • Reasoning/Modeling component = 1 point: Correct work Sample Student Response: Hector added the numerators without first finding a common denominator. $4/10 = 40/100 \text{ and } 40/100 + 5/100 = 45/100.$ Or other valid approaches are acceptable.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant



8 Rubric

1 point FIB Rubric - Part A

Score	Description
1	Student response is 8. Rationale: When 75 is divided by 9, the quotient is 8 with a remainder.
0	The response is incorrect or irrelevant.

1 point FIB Rubric - Part B

Score	Description
1	Student response is 3. Rationale: When 75 is divided by 9, the quotient is 8 with a remainder of 3. The remainder is the number of granola bars the camp counselor gets.
0	The response is incorrect or irrelevant.

15 Rubric

3 Point Constructed Response Rubric

Score	Description
3	Student response includes the following elements. • Modeling component = 1 point: Shows how to find the number of girls and boys at the Quinceañera • Modeling component = 1 point: Shows how to find the amount of pizza the girls ate and the amount of pizza the boys ate • Computation component = 1 point: Finds whether the girls or boys ate more pizza Sample Student Response:



	To find the number of girls at the Quinceañera, multiply the total number of girls and boys at the party by the fraction that are girls. $12 \times \frac{3}{4} = 9$, so there are 9 girls at the Quinceañera. To find the number of boys at the Quinceañera, multiply the total number of girls and boys at the party by the fraction that are boys. $12 \times \frac{1}{4} = 3$, so there are 3 boys at the Quinceañera. To find the amount of pizza the girls ate, multiply the number of girls by the amount of pizza each girl ate. $9 \times \frac{1}{4} = \frac{9}{4}$, so the girls ate $\frac{9}{4}$ pizzas. To find the amount of pizza the boys ate, multiply the number of boys by the amount of pizza each boy ate. $3 \times \frac{1}{2} = \frac{3}{2}$, so the boys ate $\frac{3}{2}$ pizzas. The girls ate $\frac{9}{4}$ pizzas and the boys ate $\frac{3}{2}$ pizzas, or $(3 \times 2)/(2 \times 2) = \frac{6}{4}$ pizzas. Since $9 > 6$, $\frac{9}{4} > \frac{6}{4}$. This means the girls ate more pizza. Or other valid approaches are acceptable.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

16
Rubric

2 Point Constructed Response Rubric – Part A

Score	Description
2	Student response includes the following elements. • Reasoning/Modeling component = 1 point: Valid explanation for why Jace's reasoning is incorrect. • Computation component = 1 point: Correct comparison of 24 and 12, $24=12$. Sample Student Response: Jace's reasoning is incorrect, because we need to consider both the numerator and the denominator. 24 is 2 pieces out of 4 pieces. 12 is 1 piece out of 2 pieces. They are the same because they are both half of the whole. Therefore, 24 is equal to 12. Note: the comparison statement can be given in words or symbols. Or other valid approaches are acceptable.
1	Student response includes 1 of the 2 elements.



0	Student response is incorrect or irrelevant.
2 Point Constructed Response Rubric – Part B	
Score	Description
2	Student response includes the following elements. • Reasoning/Modeling component = 1 point: Valid explanation for why Catherine's reasoning is incorrect. • Computation component = 1 point: Correct comparison of 3535 and 3737, $35 > 37$ $35 > 37$. Sample Student Response: Catherine's reasoning is incorrect because when the numerators are the same, the fraction with the smaller denominator is greater. This is because the whole is divided into fewer parts, so each part is larger. Therefore, 3535 is greater than 3737. Note: the comparison statement can be given in words or symbols. Or other valid approaches are acceptable.
1	Student response includes 1 of the 2 elements.
0	Student response is incorrect or irrelevant.

19
Rubric

3 Point Constructed Response Rubric – Part A

Score	Description
3	Student response includes the following elements. • Modeling component 1 = 1 point: Valid equation to find the number of cans of beans. • Computation component = 1 point: Correct value for the total number of cans of beans, 88



	Modeling component 2 = 1 point: Valid explanation or work shown to find the total number of cans of beans. Sample Student Response: $(3 \times 8) + (8 \times 8)$ The total number of cans of beans is 88. $3 \times 8 = 24$ $8 \times 8 = 64$ $24 + 64 = 88$ Or other valid approaches are acceptable.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

3 Point Constructed Response Rubric – Part B

Score	Description
3	Student response includes the following elements. Modeling component 1 = 1 point: Valid equation that can be used to find the total number of bags of rice. Computation component = 1 point: Correct value for the total number of bags of rice, 60 Reasoning/Modeling component = 1 point: Valid explanation or work shown to find the total number of bags of rice, Sample Student Response: $7 \times 6 + 3 \times 6$ The total number of bags of rice is 60. $7 \times 6 = 42$ $3 \times 6 = 18$ $42 + 18 = 60$ Or other valid approaches are acceptable.
2	Student response includes 2 of the 3 elements.



1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

21
Rubric

3 Point Constructed Response Rubric

Score	Description
3	Student response includes the following elements. • Modeling component = 1 point: Correct equation to find the number of \$8 shirts Olivia can buy, s • Computation component = 1 point: Correct number of \$8 shirts Olivia can buy • Reasoning component = 1 point: Correct interpretation of the remainder Sample Student Response: Olivia has \$20. She earns \$12 each hour for doing chores. She does chores for 4 hours. So, multiply 12 and 4, and add this amount to 20. Olivia wants to buy as many \$8 shirts as she can. So, divide by 8. The equation $s = (20 + 12 \times 4) \div 8$ represents the number of \$8 shirts Olivia can buy. Solve the equation. $s = (20 + 12 \times 4) \div 8$ $s = (20 + 48) \div 8$ $s = 68 \div 8$ $s = 8 \text{ R}4$ The quotient indicates that Olivia can buy 8 $\frac{4}{8}$ shirts, but it is not possible to buy part of a shirt. Olivia has enough money to buy eight \$8 shirts, but not nine \$8 shirts. Or other valid approaches are acceptable.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.